
CIS 310

Computer Organization and Assembly Language

Fall 2026 · 4 credit hours

Active Fall 2026 syllabus — Canvas provides live section details, deadlines, and submission.

Instructor	Dr. Probir Roy
Email	probirr@umich.edu
Meeting pattern	Mondays and Wednesdays; first meeting Wednesday, August 26, 2026
Class time and location	See Canvas: section, CRN, meeting time, room, and delivery mode
Office hours	See Canvas: office location, days/times, and appointment method
Course site	Fall 2026 CIS 310 Canvas
Calendar	official 2026–2027 academic calendar

Source-of-truth rule. This PDF gives the Fall 2026 course structure and the verified university calendar. Canvas controls current deadlines, points, required files, collaboration and AI-use rules, announcements, class changes, and submission receipts. If this document and Canvas differ, follow the instructor’s current Canvas announcement and ask for clarification.

Course description and prerequisites

The architecture of computer systems and associated software. Topics include digital logic circuits, computer interfacing, interrupt systems, input/output systems, memory systems, assemblers and assembly-language programming, and computer networks. This is the current UM-Dearborn catalog description.

Prerequisites: MATH 115, CIS 200, and CIS 275. Students should be ready to use binary and hexadecimal notation, Boolean reasoning, and basic programming concepts. If any prerequisite topic feels unfamiliar, use the short practice path in SystemStudio and seek help early.

Learning outcomes

By the end of the course, a successful student should be able to:

1. represent and interpret signed and unsigned data in binary and hexadecimal;
2. analyze and synthesize combinational circuits using truth tables, Boolean algebra, Karnaugh maps, adders, decoders, and multiplexers;
3. analyze and build sequential circuits using latches, flip-flops, registers, counters, and finite-state behavior;
4. explain memory organization, buses, address decoding, caches, I/O, interrupts, and the instruction cycle;
5. describe register-transfer operations, datapath/control interaction, and basic processor pipelining;
6. write, trace, and explain introductory IA-32 assembly programs, including registers, flags, memory, stack, branches, and procedures; and
7. integrate and test processor components, then communicate expected behavior, observed evidence, and remaining uncertainty.

Learning workflow and course technology

Before each topic, use SystemStudio’s short sequence **read** → **watch** → **retrieve**: read the mapped sections of the open Tarnoff text, watch the mapped official author video, and try three readiness questions without reopening the

source. Bring one unresolved point to class. Class meetings then combine explanation, prediction, worked examples, circuit or assembly demonstrations, and short evidence checks. For project work, use **predict** → **build** → **test** → **inspect evidence** → **explain**.

Canvas	The authoritative course site and the only submission destination. Confirm every due date and submission receipt there.
SystemStudio CIS 310	The VS Code extension maps all 13 lecture topics to focused open-book readings, official author videos, and readiness questions. It also provides quiz mode, confidence checks, explanations, review, the syllabus, calendar, 13 offline presentation PDFs, six assignment references, setup checks, a guided tutorial, and a local help router.
Digital v0.31	The extension can install and checksum-verify the simulator in extension-managed storage. Java 8 or newer is required for the native Digital application. Use local desktop VS Code for the graphical editor; Remote SSH is not required.
Embedded assembly lab	The extension provides Irvine32 Classroom and NASM IA-32 teaching profiles on the same bounded source-level interpreter. Windows, macOS, Linux, and Remote SSH use the same lab without Docker, Visual Studio, a native assembler/linker, or administrator access. It is a teaching subset, not full native MASM or NASM.
Student computer	A current desktop version of VS Code and enough access to install the course extension. Contact the instructor promptly if hardware, Java, accessibility, or installation creates a barrier.

The Learning Center’s reading/video checkmarks and retrieval practice are ungraded and self-reported. Its local dashboard describes attempts, accuracy, confidence mismatches, and items due for review; it does not certify mastery, predict a course grade, or send learning history to Canvas or the instructor. Use the mapped book, author video, explanation, and lecture to decide what to review, then ask for human help when uncertainty remains.

Technology recovery plan

If a tool fails, save the source file, capture the exact error and relevant screenshot, and report the expected result, observed result, evidence, and one attempted fix. Continue reading or prediction work while the issue is resolved. Do not wait until the submission deadline to report a setup problem. Canvas remains available independently of the extension.

Course materials

The extension bundles the presentation sequence locally as PDFs so students do not depend on external file-hosting access during study. The course pack also separates the three homework references from the three processor-project milestones:

- **Homework 1:** logic foundations;
- **Homework 2:** sequential logic and state machines;
- **Homework 3:** memory and assembly foundations;
- **Project 1:** registers and DRAM;
- **Project 2:** register file and ALU; and
- **Project 3:** integrated 4-bit processor.

Required open text: David Tarnoff, *Computer Organization and Design Fundamentals*. Chapter PDFs are available without charge from the [author’s official book page](#). SystemStudio links to the relevant chapter sections before each lecture; it does not redistribute the book.

Required assembly reference: Kip R. Irvine, *Assembly Language for x86 Processors*, any recent edition. Use Canvas for any updated edition or access instructions.

The mapped companion videos come from Tarnoff’s [official author channel](#) or the [ETSU open educational-resource series](#). The videos reinforce, but do not replace, the assigned reading. External book and video sites have their own privacy practices; the bundled lecture PDFs remain available locally.

Assessment, grading, and submission

Homework (3 references)	Canvas gives the released questions, points, due dates, allowed collaboration, and required submission format.
Processor projects (3 milestones)	Build and test the requested circuit artifacts. Canvas identifies individual/group rules, demonstrations, reports, and files to submit.
Quizzes and examinations	See Canvas: number, format, point values, make-up rules, and coverage.
Final examination	The university examination period is December 10–11 and December 14–16. The exact CIS 310 time and room must be taken from the Registrar/Canvas schedule, not inferred from this syllabus.
Course grade	See Canvas: assessment weights or total points, grade scale, late-work rules, incomplete requirements, and instructor-requested withdrawal policy.

Submission rule: Submit every required deliverable in Canvas and verify that Canvas recorded it. A local file, simulator run, email attachment, or SystemStudio preview is not a course submission unless the current Canvas assignment explicitly says otherwise.

Communication, feedback, and help

- Use the Canvas method specified by the instructor for course questions and private matters. **See Canvas:** the normal response-time expectation and preferred communication channel.
- Before asking about a technical problem, record: what you expected, what happened, the exact evidence, what you tried, and the decision you need help making.
- The SystemStudio Helper can route a question to a topic, tool, syllabus, calendar, or Canvas. It is local and deterministic; it does not know private grades, live deadlines, or instructor decisions.
- Seek help early when a prerequisite, setup step, circuit behavior, assembly trace, assignment instruction, or team interaction is unclear.

Academic integrity, collaboration, and AI

Students are responsible for following the [UM-Dearborn academic-integrity requirements](#), the Academic Code of Conduct, and the specific rules on each Canvas assignment. Do not submit another person's work, share solution artifacts, or misrepresent generated or copied work as your own.

Collaboration and generative-AI permissions are assignment-specific. If a Canvas assignment does not clearly authorize a type of assistance, ask the instructor before using it in submitted work. The SystemStudio workflow may organize instructor-approved materials and explain visible tool evidence, but it does not replace the student's reasoning or certify that an answer is correct.

Accessibility, well-being, and university policies

Students who encounter disability-related barriers or need academic accommodations should contact [Disability and Accessibility Services \(DAS\)](#) and communicate approved accommodations to the instructor. Course technology should not prevent equitable participation; report accessibility barriers as early as possible so an alternative path can be arranged.

Current academic policies, including attendance, grading, final examinations, and academic conduct, are linked from the University's [Academic Policies and Procedures](#) page. Student support and well-being resources remain available through UM-Dearborn. Canvas announcements identify any additional course-specific policies and updates.

Tentative Fall 2026 course calendar

The university dates below are verified against the official Fall 2026 calendar. Topic placement is a proposed instructional sequence and may move; Canvas announcements control changes. No assignment deadline is inferred here.

Mtg.	Date	Planned focus
1	Wed Aug 26	Orientation; abstraction; data representation
2	Mon Aug 31	Binary and hexadecimal representation
3	Wed Sep 2	Signed data, two's complement, and adders
–	Mon Sep 7	Labor Day holiday — no class
4	Wed Sep 9	Boolean operations and truth tables
5	Mon Sep 14	Boolean algebra and circuit simplification
6	Wed Sep 16	Karnaugh-map method
7	Mon Sep 21	Karnaugh maps and don't-care conditions
8	Wed Sep 23	Decoders, multiplexers, and data selection
9	Mon Sep 28	Combinational design and evidence-based testing
10	Wed Sep 30	Latches, clocks, and state
11	Mon Oct 5	Flip-flops, registers, and counters
12	Wed Oct 7	Sequential-circuit analysis and verification
13	Mon Oct 12	Memory organization and memory maps
14	Wed Oct 14	Buses and address decoding
15	Mon Oct 19	I/O protocols and polling
16	Wed Oct 21	Interrupts and asynchronous behavior
17	Mon Oct 26	Memory hierarchy and storage latency
18	Wed Oct 28	Cache and locality
19	Mon Nov 2	Register-transfer language and micro-operations
20	Wed Nov 4	Arithmetic units and ALU design
21	Mon Nov 9	Control unit and instruction cycle
22	Wed Nov 11	Datapath/control integration
23	Mon Nov 16	Processor components and pipelining
24	Wed Nov 18	Address spaces and x86 registers
–	Nov 21–29	Thanksgiving recess — no class Nov 23 or Nov 25
25	Mon Nov 30	Assembly syntax, translation, and execution model
26	Wed Dec 2	Embedded assembly tracing: registers, flags, stack, and memory
27	Mon Dec 7	Integration, review, and last regular class
–	Dec 8–9	Study days
–	Dec 10–11, 14–16	University examination period; verify the CIS 310 slot in Canvas

Keeping course information current

Use this syllabus for the stable course structure and the SystemStudio calendar for official term dates. Use Canvas for section-specific details and anything that can change during the semester: meeting room or delivery updates, office hours, updated Irvine edition/access instructions, assignment and examination instructions, grading rules, deadlines, collaboration/AI permissions, and the final-exam time and room. When two sources appear inconsistent, follow the newest instructor announcement in Canvas and ask for clarification.

Source note. Updated August 19, 2026. The current UM-Dearborn catalog supplies the course title, credits, description, and prerequisites. The official UM-Dearborn 2026–2027 academic calendar supplies term dates. The prior course syllabus supplies the required texts; Tarnoff's official pages supply the open chapters and author videos. The locally packaged Fall 2026 course materials supply the topic sequence. Canvas is authoritative for live section details and submission.